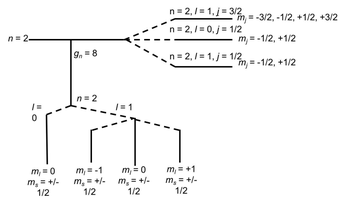
Problem 4.13 - January 11, 2021

JD **John W. Daily** **Verified Instructor** 4 years ago

Consider the energy levels of a hydrogenic atom for which S=S=3. In terms of appropriate quantum number values, list the different eigenstates, first assuming no spin-orbit coupling and then with spin-orbit coupling. Compare the degeneracies for the two cases (including electron spin) and sketch the component levels that result from spin-orbit coupling. Use Fig. 4.13 as a guide.



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PL **Peter Linder** **Learner** 4 years ago

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KK **Khairul Kabir** **Learner** 4 years ago

When the familiar red spectral line of the hydrogen spectrum is examined at very high resolution, it is found to be a closely-spaced doublet. This splitting is called fine structure and was one of the first experimental evidences for electron spin.

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KK **Kalaivanan K** **Learner** 4 years ago

When the familiar red spectral line of the hydrogen spectrum is examined at very high resolution, it is found to be a closely-spaced doublet. This splitting is called fine structure and was one of the first experimental evidences for electron spin. The small splitting of the spectral line is attributed to an interaction between the electron spin S and the orbital angular momentum L. It is called the spin-orbit interaction.

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